

DHYANI SONI

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Education

New Jersey Institute of Technology

Expected Graduation Date

B.S. in Computer Science

May 2027

- Albert Dorman Honors College | GPA: 3.84 | Dean's List | Recipient of Highlander Achievement Scholarship
- **Relevant Coursework:** Data Structures, Data Science, Linear Algebra, Database Management, Artificial Intelligence, Internet Applications, Operating Systems, Computer Networks

Technical Skills

Languages: Java, Python, SQL, JavaScript, PHP, C/C++

Frameworks & Libraries: Pandas, NumPy, Scikit-learn, PyTorch, OpenCV, FastAPI, React, Spring Boot, Streamlit

Tools & Platforms: VS Code, AWS (S3, RDS), Firebase, PostgreSQL, Jupyter Notebook, Git, Tableau, Postman, Anaconda, FAISS

Experience

Clinical Neuromuscular Adaptation Laboratory

Newark, NJ

AI Research Assistant @Department of Biomedical Engineering

Sep 2025 – Present

- Applied **computer vision** and **machine learning** to extract movement patterns from noisy underwater data, enabling more accurate **analysis** of neuromuscular properties.
- Integrated **open-source pose estimation algorithms** to improve body movement tracking in underwater environments, supporting clinical research goals.
- Designed **preprocessing** pipelines with underwater filtering to generate clearer images and establish **ground truth** color references, **improving dataset reliability**.

Department of Informatics - NJIT

Newark, NJ

Research Assistant

Apr 2024 – Aug 2025

- Conducted **literature review** on accessibility for older adults; **designed** a mixed-methods study addressing online safety behaviors.
- Collected and analyzed data from **25+ semi-structured interviews** using **Otter ai**, created **storyboards** of safe interactions, and generated inclusive design recommendations.
- Presented findings in lab meetings, translating technical results into actionable insights for faculty and peers.

Projects

NoteShare | Spring Boot, React, PostgreSQL, AWS (S3, RDS), JWT

Aug 2025

- Built a full-stack platform for peers to **upload, search, and download notes** using React and **Spring Boot**.
- Secured access with **JWT** authentication and **role-based authorization** for **students** and **admins**.
- Integrated **AWS S3** and **RDS** for scalable, secure file and metadata storage for a **cloud-optimized solution**.

DermaTag | CNN, FastAPI, React, Canva Code

July 2025

- Developed **full-stack AI skincare tool** using **CNN (PyTorch)** and **FastAPI** to classify facial skin concerns from webcam images, improving product matching accuracy
- Integrated **image-based inference** and **tag-based recommendation logic** with real product links
- **Live camera capture** feature gave **real-time predictions**, and **personalized** recommendations, increasing skincare website engagement

AgentWeave | Python, FastAPI, YOLO, FAISS, Firebase, React

May 2025

- Programmed a full-stack **AI outfit recommender** using **CLIP embeddings**, **BLIP captions**, and **FAISS vector search**.
- Integrated **YOLOv5** to **detect body shape** and **personalize** outfit recommendations.
- Built a React interface with **Tinder-like** swipe gestures connected to Firebase for real-time feedback.

Leadership & Activities

Lyra STEM, Curriculum Manager: Designed and implemented STEM curricula for high school students, enhancing hands-on learning

Kids Who Code, Treasurer: Managed budget and fundraising activities, ensuring efficient allocation of resources

Rewriting the Code, Member: Attended technical workshops and collaborative coding sessions

SHPE, Member: Participated in STEM outreach events and professional development activities

CodePath, Student: Completed training in SWE interview preparation, data structures and algorithms